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A Distance-Based Scheduling Algorithm with a Proactive Bottleneck Removal Mechanism for Wireless Rechargeable Sensor Networks

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ABSTRACT The development of wireless charging technology has made research in rechargeable wireless sensing networks (WRSNs) a hot topic. Charge scheduling, which involves allocating mobile charger(s) to replenish energy to the sensors of a network so that the network can remain active for a longer time, is one of the most essential issues in this field. Most previous works addressed the charge-scheduling problem by using proactive or on-demand methods, each with its own advantages and issues. In this work, we propose a novel *bottleneck prediction and removal mechanism* (BP&R) that seamlessly integrates the two completely different scheduling strategies. The BP&R predicts proactively the potential charging needs that may overwhelm the on-demand scheduling and eliminate those incidents beforehand. This allows the on-demand scheduling, which usually relies on time and/or distance factors in determining charging order, to concentrate on tackling the problem considering only the distance factor. A simple distance-based near job first (NJF) algorithm is thus sufficient for charge scheduling when time-related factors are considered prior to the determination of the charging order. Use of NJF algorithm also allow us derive a shorter path length in charge scheduling, which results in a tighter charging threshold. A lower charging threshold corresponds to fewer charging requests and less pressure on charge scheduling. In general, the new scheduling strategy relies on the on-demand feature to gather the charging demands of the network, while proactively includes additional sensor nodes that could potentially cause subsequent issues in the scheduling plan. The on-demand and proactive natures of the proposed strategy ensure that it can adapt to the dynamic charging needs of sensors, while at the same time taking precautionary measures to avoid possible burst requests in the future. Extensive simulations show that the proposed strategy can achieve a better network lifetime than other state-of-art scheduling methods.

INDEX TERMS wireless rechargeable sensor networks; charge scheduling; semi-proactive charging algorithm; scheduling bottleneck prediction and removal; sustainable networks

I. INTRODUCTION

Wireless sensor networks (WSNs) consist of a large number of inexpensive miniature sensors deployed in specific areas for event monitoring. The functions of wireless sensor networks include sensor collaboration awareness, information collection, and target monitoring. Potential applications include military, aviation, disaster relief, environmental, medical, industrial, and other fields. Energy is critical for WSNs because their sensor nodes are usually powered by on-board batteries or super capacitors. Due to

the small size of the sensor nodes, their battery capacity is limited. Sometimes it is not possible to extend the life of wireless sensors by replacing their batteries due to high maintenance costs, which limits the development and application of WSNs [1].

Recently, progress in wireless energy transmission (WPT) [2] has resolved the energy supply issue of sensor nodes. In [2], Kurs demonstrated the feasibility of an efficient energy supply over a distance through coupled resonance. Kurs also developed a series of prototype devices for wireless energy

transmission [3]. This technology provides new ideas for wireless sensor networks. Previous research suggested that wireless chargers equipped with coupled coils could use this new technology to charge sensor nodes at a close distance. A vehicle can carry these chargers and move around a WSN to provide a stable energy supply to the nodes in the network. With proper planning of the traveling path of this so-called *wireless-charging vehicle* (WCV), it is feasible to promptly recharge the nodes in a WSN and thus extend the service life of the WSN. The aforementioned WCV charging path planning corresponds to the charge-scheduling problem. Most previous work tackled this charge scheduling issue by using proactive or on-demand methods [4], [5].

The proactive method assumes that the WCV (or *base station*, denoted BS hereafter) records some necessary information on the sensor nodes (such as the position coordinates and energy state). The WCV or BS can then make a scheduling plan in advance to charge the network (usually in a periodic routine). Proactive (or periodic) methods may work well if sensors in *wireless rechargeable sensor networks* (WRSNs) operate with stable power consumption rate, however, it may not adapt well to the dynamic change of network operations. For example, some particular events may result in unexpected power draining from certain nodes prior to their scheduled charging times. This issue with the proactive method may not be resolved unless there is a way to determine in real-time the power consumption of each sensor node and change the pre-determined charging order accordingly.

The on-demand method operates by requiring that any sensor node actively monitors its remaining energy and sends a charging request to the BS (or WCV) when its energy level drops below a certain threshold. The BS (or WCV) then maintains a service pool to store the received requests and engages in charge scheduling accordingly. Unlike its counterpart, the on-demand method does determine the distribution of charging demands before acting upon the requests. However, if there are too many charging requests in a short period, a potential bottleneck may make the WCV fail to meet all its charging requests in time. This may be a potential disadvantage of the on-demand method.

Two important issues with on-demand scheduling are determining the charging threshold and the charging order of the to-be-charged sensors. Although a threshold that is too loose or too tight may have a negative effect on WRSN [6], [7], previous works usually set this threshold as 15%~40% of a sensor's battery power [8] - [12] without giving details regarding their choices. In addition, most previous on-demand scheduling methods consider time (the deadline for a sensor to become recharged), distance (the travel distance between the WCV and the to-be-charged sensor), or both factors combined through some weights to calculate the priority of the charging tasks [8], [13] - [15]. Estimation of the charging threshold involves more accurate knowledge of the average travel distance of the WCV. To make this matter even more complicated, the threshold value also affects the deadline of

the sensors. In other words, the charging threshold, time, and distance factors are correlated and should be considered in the same context (unlike the approaches of previous studies).

The aforementioned factors motivated us to develop a completely new charge-scheduling algorithm. In this work, we propose a *bottleneck prediction and removal mechanism* (BP&R) that seamlessly integrates the two completely different scheduling strategies discussed above. The proposed novel scheduling strategy relies on the on-demand feature to roughly determine the power consumption in a network. While reacting to the on-demand charging requests, the proposed method also includes proactive sensor nodes, selected by the bottleneck prediction mechanism, which could potentially cause a charging bottleneck in the future as long as there is spare WCV battery power and time. One advantage of applying BP&R is that we can predict, over time, the potential charging needs that may overwhelm the on-demand scheduling and eliminate those incidents beforehand. This is similar to pulling the time factor from on-demand scheduling and putting it into a more global context to concentrate on tackling the scheduling problem considering only the distance factor. In particular, all the charging tasks, both those that issue the charging request actively and those that are proactively selected, are placed in the service pool and are then scheduled with a simple distance-based near job first (NJF) algorithm. The on-demand and proactive features of the proposed strategy ensure that this method can adapt to the dynamic charging needs of sensors, while at the same time taking precautionary measures to avoid possible burst requests at a later time.

Another advantage of removing the time factor is that we are able to derive a shorter path length of charge scheduling based on the NJF algorithm, which results in a tighter charging threshold. Accordingly, in this work, we first develop mathematical models to evaluate the upper bound of the path length using the NJF scheduling algorithm. We then use the models to determine the appropriate threshold of battery power that a sensor needs to issue its charging request. This charging threshold is much lower than that used in earlier research yet may still work well with our proposed algorithm without compromising the network's lifetime. In addition, the same models are also used to estimate the size of the service pool, which corresponds to the number of charging tasks that a WCV may accomplish on a charging mission.

We then conduct extensive simulations to evaluate the performance of the proposed charging-scheduling mechanism. The simulation results demonstrate that the proposed strategy, when combining NJF or TADP [8] with the developed BP&R mechanism, can improve the network lifetime by up to 210.6% and 166.8%.

The rest of this paper is organized as follows. Section II presents the relevant studies, and Section III provides the network environment used in this work. Section IV describes the details of the proposed method, with the simulation results

of the proposed mechanisms presented in Section V. Finally, Section VI summarizes this paper.

II. RELATED WORK

The essential technology that makes wireless rechargeable sensor networks possible is the wireless power transfer (WPT) mechanism. Depending on the power transfer range, the WPT can be categorized as near field [16] or far field [17]. The transfer of power can be done through electromagnetic radiation [18], resonant coupling [19], or inductive coupling [20]. The antenna used can be omnidirectional (OWPT) [21] or directional (DWPT) [22], [23]. The former generates uniform radio frequency (RF) signals in all directions, which may result in lower power transfer efficiency. On the other hand, directional WPT technology uses directional antennas or antenna arrays to generate concentrated energy beams and directs the energy beams to selected areas, thereby reducing the disadvantages of omnidirectional technology. In addition, according to [3], some studies also consider a setup wherein a single mobile charger provides a power transfer to multiple sensors simultaneously to speed up the charging process [6], [24], [25].

Many related studies proposed solving the problem of scheduling mobile charger(s) so that energy can be provided to sensors in wireless sensor networks. These methods can be roughly categorized into two types: proactive (or periodic) and on-demand charging.

In terms of proactive charging, the WCV periodically accesses sensors in the network and charges the sensors regardless of whether the sensors need immediate attention. This method is usually more suitable in networks with relatively stable sensor energy consumption, for which optimization of the charging path is performed. Since the optimization problem is NP-HARD [26], many studies use heuristic approaches (like genetic algorithms or similar methods) to solve this problem. Lyu et al. [26] proposed a Hybrid Particle Swarm Optimization Genetic Algorithm (HPSOGA) for charging path planning with the assumption that the power consumption rate of the sensors is constant. Zhao et al. [27] proposed a scheduling algorithm that uses the Teaching and Learn based Optimization method to find the best charging path according to periodic requirements.

Based on the energy allocation and consumption model, the proactive charging scheme [28] converts the energy charging scheduling problem into a passenger salesman problem and calculates the shortest Hamiltonian period as the solution. Charging schemes can be further differentiated based on the number of sensors that the WCV can charge at a time. In the single-sensor charging scheme [29], only one sensor can be charged at a time. With a multi-sensor charging scheme [30], multiple sensors within the charging range can all be charged at the same time to improve charging efficiency. However, due to the uncertainty of environmental changes, the energy consumption of the sensors in a WRSN may have dynamic characteristics, which makes it difficult for the periodic

charging method to catch up. Therefore, an on-demand charging mechanism may be more suitable for the aforementioned complex and dynamic environment.

On-demand charging methods usually rely on the sensor's self-monitoring mechanism. When the remaining energy of a sensor drops below a predetermined threshold, the sensor sends a charging request to the BS (or the WCV). The BS/WCV then arranges the charging sequence of the sensors based on these collected requests with certain scheduling rules. One of the most notable scheduling rules is Earliest Deadline First (EDF), which only considers task deadlines to determine the order of charging task scheduling [5]. In other words, the task closest to the deadline is always selected as the next recharging target.

He et al. [14] proposed a preemptive "Nearest Job Next" (NJNP) method, which is a typical on-demand charging scheme. For NJNP, the WCV always selects the nearest sensor from the waiting list as the next charging sensor. The disadvantage of the NJNP algorithm is that it does not consider the remaining energy of the sensor. In other words, a half-charged sensor located closer to the WCV may have higher charging priority than another more distant and dying sensor.

Shu et al. [31] used speed control of the WCV to reduce the total moving time of the WCV and increase the effective charging time of the sensors, which is a key design issue for WRSNs [31]. The authors first proved that the speed control in the given problem was NP hard and then developed a heuristic solution with a provable upper bound. A large number of simulations show that by using this proposed speed control mechanism, the lifetime of the WSN network can be extended by 250%.

Lin et al. [8] designed a time-space priority (TADP) scheduling algorithm. This algorithm considers the remaining work time and distance of the sensors issuing charging requests. These two factors are sorted separately and the sorted order is combined into mixed priorities. Extensive simulations were performed to demonstrate the advantages of the TADP method. Zhong et al. [15] argued that, instead of the remaining work time, the energy consumption rate of sensors should be considered since sensors with higher energy consumption rates may die faster. Tomar et al. [6], [25] used fuzzy logic to fuse network information, such as residual energy, distance to the WCV, critical node density, and energy consumption rate as the basis of the charging schedule.

Lin et al. [9] proposed a primary and passer-by scheduling scheme for large WRSNs based on an on-demand charging architecture. The algorithm first uses the deadline to sort the charging requests issued by the sensors. Each time when dispatching the WCV, ten charging tasks with the earliest deadline are taken to create the so-called primary order and then the consecutive nodes in this primary order are searched for passer-by charging requests that do not affect the charging mission. The selected charging requests are then added to the primary charging path to improve charging efficiency.

Lin et al. [10] proposed an optimal path planning and charging scheme for on-demand charging architecture. The program determines whether a feasible schedule can be found to complete all charging tasks. If one is available, the algorithm provides the best charging path to maximize charging efficiency. Otherwise, a scheme is proposed to discard some of the requests.

Xu et al. [4] proposed a new model using a WCV and multiple detachable chargers. The WCV can leave the charger on the sensor node to perform charging tasks. Based on this new model, a revised EDF (REDF) scheduling algorithm was proposed. A large number of simulation results show that, in terms of the service life of the WRSN, the number of successful charging requests, and the average moving distance of the WCV, the performance of REDF is better than that of EDF.

Wang et al. [32] proposed a new model combining on-demand charging and a data collection model (CRCM). The proposed model collects data and simultaneously charges the sensors. The authors designed a hexagon-based algorithm to classify all sensor nodes into different clusters and then use the remaining energy and geographic location (REGP) algorithm to calculate the priority of each cluster. Simulations show that the performance of REGP is better than that of EDF and NJNP.

Kaswan et al. [11] proposed a WRSN linear programming formula using a single WCV on-demand architecture. The goal of their solution was to minimize the charging delay of the WRSN. To solve the charging scheduling problem effectively, the authors proposed a gravity search algorithm (GSA). This method considers both time and space. The simulation results show that the GSA-based solution performs better than the "first come, first served" and NJNP methods in terms of charging latency.

Lin et al. [12] proposed a double warning threshold and double preemption (DWDP) charging strategy. Whenever the sensor's remaining energy level falls below the two predetermined thresholds, two warnings are issued separately. With the early warning, the scheduling of charging tasks can have better flexibility and performance.

Zhao et al. [27] proposed a scheduling algorithm that first uses the Teaching and Learn based Optimization method to find the best charging path and the appropriate charging time for each request according to periodic requirements. Dynamic (on-demand) requests are inserted upon demand into the previously derived periodical schedule. This hybrid scheme can improve the charging efficiency and charging success rate by nearly 100%.

In addition to studies on WPT and charge scheduling, a seemingly neglected issue is determining the threshold value for sensors to issue the charging requests in an on-demand scheduling context. A charging threshold that is too loose or tight may have negative effects on the WRSN [6], [7]. However, previous studies usually set a fixed threshold value [8] - [12] without justifying this value. Indeed, estimating the charging threshold involves knowledge on the travel distance

of the WCV. Some works use the average distance between any two points in the network to estimate the moving distance of the WCV [6], [7], which may result in a threshold that is too loose.

In this work, we propose a hybrid scheduling method that combines the features of both on-demand and proactive schemes. This method acts according to an on-demand charging mechanism by first responding to the charging requests of the sensors. In the meantime, when the BS produces the charge scheduling plan, it also proactively includes some sensor node that do not need immediate attention yet may potentially cause issues in the future, as long as the WCV has spare time and battery power. The on-demand and proactive features of the proposed algorithm ensure that it can adapt to the dynamic charging needs of sensors, while at the same time taking precautionary measures to avoid possible request bursts at a later time. In addition, this paper also discusses the determination of an appropriate threshold for any given sensor to issue its charging request.

III. NETWORK ARCHITECTURE

In this section, we describe the architecture and components of the WRSN model that our proposed novel scheduling algorithm operates upon, followed by notations and terminology used throughout this paper.

The WRSN architecture, which consists of three main components, including a BS, a WCV, and a number of sensors, is shown in Fig. 1. Detailed descriptions of each component are presented as follows.

- (1) A BS fixed facility located at the bottom-left corner of the sensor network area and equipped with many super-capacity batteries. It is capable of quickly replacing the battery of the WCV and obtaining the location of all sensors through a one-time collection. This work assumes that the BS does not know the current residual energy of all sensors. Instead, the BS maintains a table *S*Table that records the estimated remaining work time of each sensor. When each sensor is activated, the sensor sends an activation message to the BS, including its estimated remaining work time. The BS will then set the estimated remaining work time of the sensors in the *S*Table accordingly. In addition, when the WCV completes a charging mission and returns to the BS, it also updates the *S*Table based on the collected information from the sensors that it charged. Moreover, the BS maintains a queue for charging requests. When the WCV is at the BS and the queue is not empty, the BS will schedule the requests in the queue and notify the WCV to start a charging mission.
- (2) A WCV with a single power system that can move by itself and charge the sensors. The WCV needs to reach the target sensor's position to provide one-on-one charging. When notified by the BS, the WCV will move to and fully charge each sensor, as scheduled by the BS. To simplify the system architecture, the WCV only receives the

charging instruction at the BS and does not return to the BS until all charging tasks are completed. Once the WCV returns to the BS, its battery can be replaced immediately and be readied for its next charging mission.

- (3) A set of homogeneous rechargeable sensors, each one aware of its location and able to stop working when its energy is exhausted. These sensors are randomly deployed to monitor the environment of interest. Each sensor actively keeps track of its remaining energy and issue a charging request to the BS when the energy level drops below a predefined threshold ϕ . We do not consider the delivery time of the charging request, as it is negligible when compared to the WCV travel and sensor charging times.

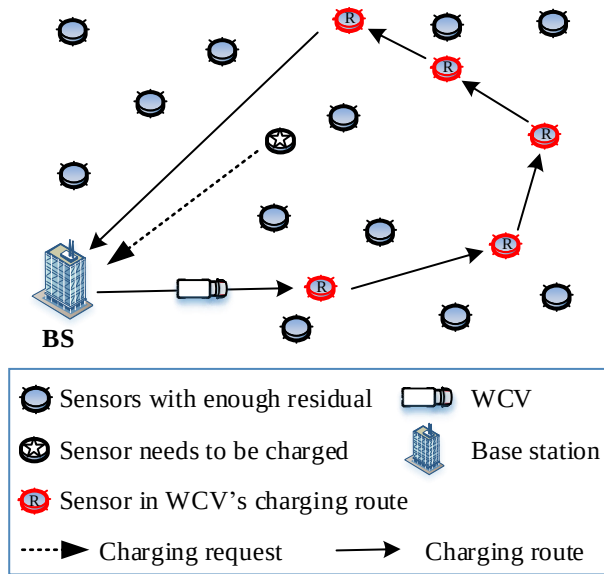


Figure 1. The WRSN architecture of our proposed algorithm

Next, we summarize the symbols and definitions used throughout this paper in Table I. As described earlier, this work assumes that the WCV is alone once it leaves the BS. In other words, the EDF, TADP, or NJF scheduling discussed in this paper work without a preemptive mechanism.

TABLE I
SYMBOLS AND DEFINITIONS

Symbol	Definition
$N_{maxTask}$	Number of tasks a WCV can handle on a charging mission
N_{on_demand}	Number of on-demand charging tasks
$N_{proactive}$	Number of proactive charging tasks
N_{list}	Number of elements in the list
L_{path}^k	Distance for a WCV's charging mission, i.e., the distance for a WCV to move from the BS, passing through k sensors, and back to the BS.
$L_{maxStep}$	Maximum distance between any two points in the network area.
$T_{job}(k)$	Turnaround time for a WCV's charging mission, i.e., the time for a WCV to complete k charging tasks and return to the BS
T_{req}	Threshold of the remaining work time for a sensor to issue a charging request
T_{full}	Time to fully charge a sensor from 0% to 100%

$T_{Current}$	Current time
$T_{maxMove}$	Maximum time for a WCV to move between any two points in the network area
$T_{deadline,x}$	The estimated deadline for sensor x to issue the charging request
$T_{deadlineThreshold}$	The threshold value for determining the tolerant error range of the estimated deadlines
E_{car}	Energy capacity of the WCV
E_{sensor}	Energy capacity of the sensor
E_{move}	Energy consumption for moving the WCV
$E_{job}(k)$	Energy consumption for a WCV's charging mission, i.e., the energy required for the WCV to complete k tasks and return to the BS
$V_{charging}$	Charging rate of the sensor from the WCV
V_{car}	Traveling speed of the WCV
F_{req}	Frequency of the charging request issued in the network

IV. PROPOSED SCHEME

Time and distance are two factors that are commonly adopted to determine the charging schedules of wireless rechargeable sensor networks. However, there is always a trade-off in choosing one over the other or determining their weighting when both are considered. The EDF [5] method considers the deadlines of the charging requests only and selects the charging request with the nearest deadline for charging. However, no consideration is given to the positions of the sensors, which results in a longer WCV travelling distance. NJF [14] selects the nearest charging request for recharge each time and saves the WCV from moving a long distance. However, considering only distance may cause farther sensors to die due to not being charged for a long time. TADP [8] considers both time and distance factors simultaneously and determines the charging order with mixed time / distance weights. However, the inclusion of a temporal factor can still cause the WCV to move a longer distance and result in the loss of time and energy.

As long as the charging tasks can be completed in time, the temporal factor does not need to be taken into account in each schedule. Therefore, this work will use mainly the NJF method for scheduling to obtain a schedule with a short WCV moving distance while taking into consideration the time factor based on the following aspects:

- (1) Determine the environment in which the NJF can be used safely—that is, based on factors such as the size of the network, the charging rate of the sensor, the traveling speed and energy consumption of the WCV, the possible frequency of charging requests, and so on, to determine whether the charging needs can all be fulfilled in time using NJF. In Section IV.A and Section IV.B, we present more discussions related to this issue.
- (2) Activate the proactive charging mechanism in case there are future possibilities that NJF will not be able to handle the accumulated charging requests in time. The idea of the proactive charging mechanism proposed in this paper is to foresee potential charging bottlenecks and utilize leftover power and time (in addition to completing the on-demand charging requests) of the WCV in each of its missions to charge the selected sensor(s) based on charging needs within the predicted bottleneck period. In

this way, we can discriminately select some sensors and fulfill their charging requirements in advance to avoid potential bursts of charging requests later. In addition, as the number of charging tasks increases (because of additional proactive tasks), there is a better chance of obtaining more optimized scheduling. Fig. 2 shows the proposed scheme with more details, including the mathematical basis, which is presented in the following subsections.

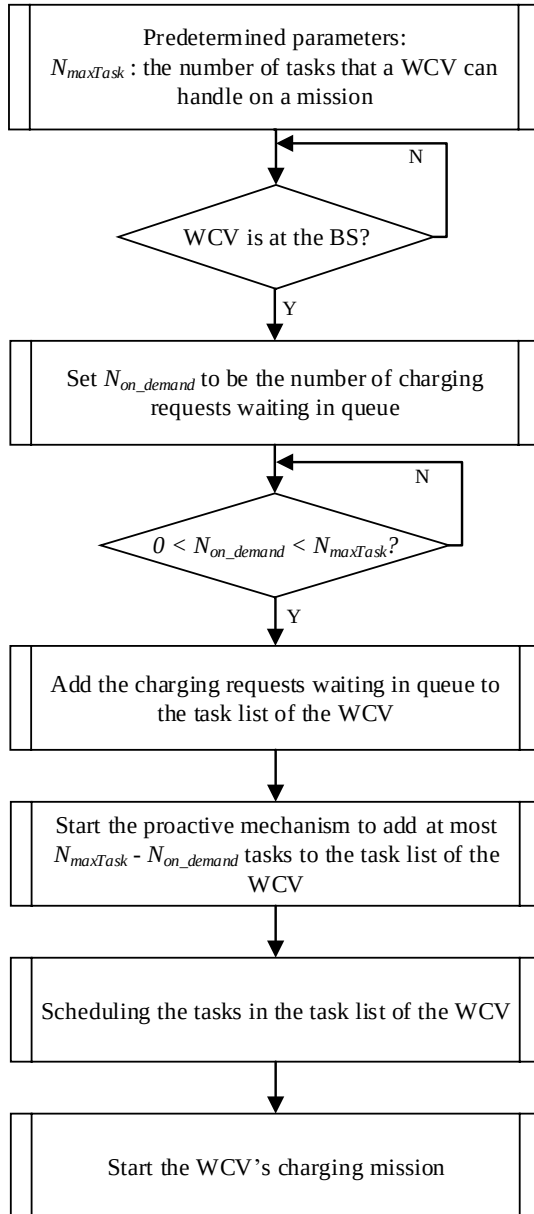


Figure 2. The proposed charging scheduling scheme with a proactive mechanism.

A. MATHEMATICAL MODEL FOR THE THRESHOLD VALUE TO ISSUE A CHARGING REQUEST

In some earlier works, the average distance between any two points in the network was used to estimate the moving distance

of the WCV [6], [7]; this method usually results in a threshold that is too loose. In the following, we develop mathematical models to evaluate the path length of charge scheduling based on the nearest job first algorithm. The models are then used to estimate the number of charging tasks that a WCV may accomplish on a charging mission, as well as evaluate the appropriate threshold of battery power (T_{req}) that a sensor needs to issue its charging request.

The longest distance between any two points in a square network area with side length a is

$$L_{maxStep} = \sqrt{2}a. \quad (1)$$

Thus, the maximum time required to move between any two points is

$$T_{maxMove} = L_{maxStep} / V_{car}. \quad (2)$$

Since the WCV needs to return to the BS after completing its charging tasks, the turnaround time required to complete k charging tasks is

$$T_{job}(k) = \frac{L_{path}^k}{V_{car}} + k \times T_{full} \text{ where } T_{full} = \frac{E_{sensor}}{V_{charging}}, \quad (3)$$

and the energy required to complete k charging tasks is

$$E_{job}(k) = L_{path}^k \times E_{move} + k \times E_{sensor}. \quad (4)$$

In addition, the frequency F_{req} of the charging requests generated in the network should be less than the frequency with which the WCV completes its tasks. Otherwise, there will inevitably be charging tasks that fail to be completed in time. That is,

$$F_{req} < \frac{k}{T_{job}(k)}. \quad (5)$$

It is assumed that the frequency of the charging request issued is based on Equation (5)—that is, a maximum of $N_{maxTask}$ requests occur within $T_{job}(N_{maxTask})$ time, considering a scenario where a charging request x is issued exactly when the WCV leaves the BS for its charging mission. Upon the WCV completing its current mission and returning to the BS, there may be at most another $N_{maxTask} - 1$ pending charging request. In the worst case, request x may be scheduled last (the $N_{maxTask}$ -th charging node) on the next mission. Thus, the maximum waiting time for request x is

$$T_{job}(N_{maxTask}) + \frac{L_{path}^{N_{maxTask}} - L_{maxStep}}{V_{car}} + (N_{maxTask} - 1) \times \frac{E_{sensor}}{V_{charging}}. \quad (6)$$

Therefore, the threshold for issuing the charging request, T_{req} , of a sensor should meet the following condition:

$$T_{req} \geq T_{job}(N_{maxTask}) + \frac{L_{path}^{N_{maxTask}} - L_{maxStep}}{V_{car}} + (N_{maxTask} - 1) \times \frac{E_{sensor}}{V_{charging}}. \quad (7)$$

Eq. (7) gives the lower bound of the charging threshold that a sensor needs to send its charging request. Clearly, instead of using the average of any two points in the entire network to estimate the distance traveled by a charging vehicle [6], [7], a more accurate approximation of L_{path}^k would correspond to a more appropriate charging threshold setting, T_{req} . We formally

discuss the model for path estimation in the following subsection.

B. MODEL FOR NJF PATH LENGTH ESTIMATION

Based upon our observations, NJF-like algorithms perform well when the assigned charging tasks have looser deadlines. On the other hand, when the deadlines are tighter, EDF-like algorithms usually perform better than other algorithms. As a result, NJF-like and EDF-like algorithms are considered the baselines of performance for the newly proposed charging scheduling algorithm.

In this subsection, we estimate the longest distance of the path starting from the BS, passing through k sensors, and back to the BS, denoted as L_{path}^k and defined in Table I for the NJF algorithm. The derived value of L_{path}^k can be used to determine the threshold, T_{req} , which may affect the proposed proactive charging strategy significantly.

Suppose the deployed network is an $a \times a$ square area. Since the longest distance between any two points in a square with a side length a is $\sqrt{2}a$, we have a trivial upper bound of

$$L_{path}^k \leq (k+1) \times \sqrt{2}a. \quad (8)$$

However, Jaillet [33] proposed a tighter upper bound of the shortest path, which is

$$L_{path}^k \leq \sqrt{2}ka + 4a. \quad (9)$$

In this work, we propose a new tighter bound of the value L_{path}^k for the NJF algorithm.

When applying NJF, assuming that the WCV is currently at point x , if y is selected as the next stop, there is no other point in the circle with x as the center and the distance between x and y as the radius. Based on this characteristic, there should be a closer approximation of the longest path length estimated for NJF. The intuitive idea in the proposed new bound is described as follows. Since the path with the longest distance often occurs in symmetrical and extreme cases, we thus consider scenarios where the BS and sensors are dispersed in grid points. As an example, one of the grid deployed scenarios of the four sensors, when $k=4$, is shown in Fig. 3a. Other cases for $k=9$ and $k=16$ are shown in Fig. 3b,c, respectively. In these cases, as shown in Fig. 3, the NJF paths are $3a$, $4a$, and $5a$ for $k = 4, 9$, and 16 , respectively.

Based on this observation, we use $(k-1) \times a / (\sqrt{k} - 1)$ to estimate the longest length of the NJF path containing k points. Finally, the maximum distance from the BS to the start of the path and the distance from the end of the path back to the BS is $2\sqrt{2}a$. Accordingly, we obtain

$$L_{path}^k \leq (k-1) \times \frac{a}{\sqrt{k}-1} + 2\sqrt{2}a. \quad (10)$$

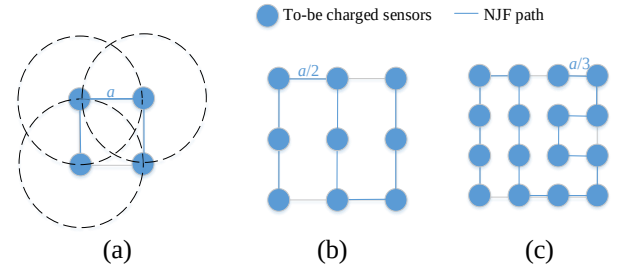


Figure 3. Examples of the longest NJF path when sensors are deployed in grid points. (a) 4 sensors, (b) 9 sensors, and (c) 16 sensors.

NJF-like algorithms choose the nearest job as the next job to be served. Thus, in the following theorems, the nearest point first uses the same concept as NJF—that is, choosing the nearest point as the next point to visit. The following theorems give formal proofs for Equation (10) when $k=2, 3$, and 4 .

Theorem 1. In a square network area with side length a , two points x, y are arbitrarily selected. Let L_{path}^2 be the path length starting from the base station and returning to the base station after passing the x and y points in the order of the nearest point first. Assuming that the BS is located in the center of the network area, then $L_{path}^2 \leq 2\sqrt{2}a$.

Proof of Theorem 1. The longest distance between any two points in a square area with side length a is equal to $\sqrt{2}a$, and the longest distance from any point in the area to the central BS is $\frac{\sqrt{2}}{2}a$; thus, $L_{path}^2 \leq \sqrt{2}a + 2 \times \frac{\sqrt{2}}{2}a = 2\sqrt{2}a$. ■

Note that this value is less than the value $(\sqrt{2} + 1)a + 2\sqrt{2}a$ obtained by Equation (10).

Theorem 2. In a square network area with side length a , three points x, y, z are arbitrarily selected. Let L_{path}^3 be the path length starting from the BS and returning to the BS after passing points x, y and z , in the nearest point first order. Assuming that the BS is located in the center of the network area, then $L_{path}^3 \leq a + 2\sqrt{2}a$.

Proof of Theorem 2. First, we prove that $a + \sqrt{2}a$ is the sum of the maximum distance formed by any three points in the order of the nearest point first. In general, there are only two cases to consider.

Case 1: The three points are all on the corners. Choose three points from the four corner points of the square, and the longest total distance is $a + \sqrt{2}a$ from all combinations.

Case 2: The three points are not necessarily on the corners. Without a loss of generality (due to the symmetry of the network area), we only need to discuss the cases where the starting points are within the red triangle area in the figure below (Fig. 4).

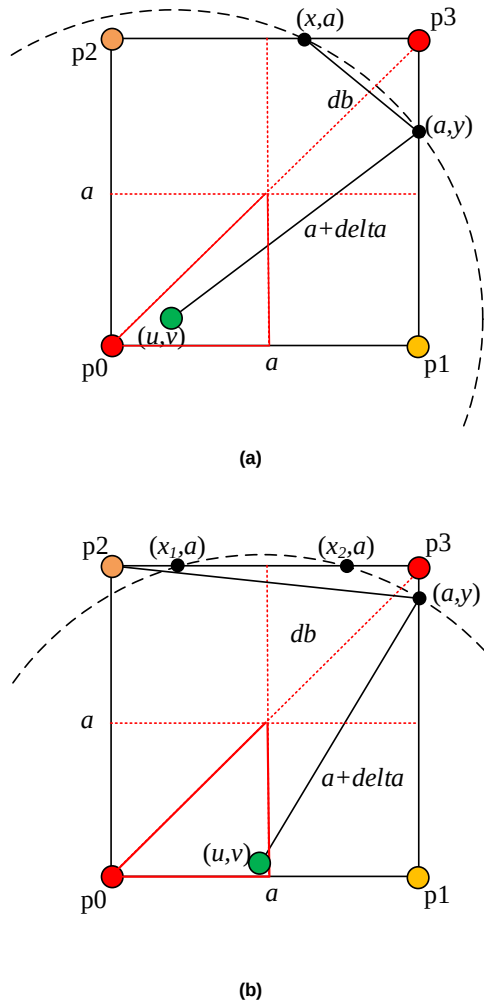


Figure 4. Two path scenarios between the first point (u, v) and point p_2 . (a) less than $a + \delta$, (b) greater than or equal to $a + \delta$.

Suppose that p_0 is used as the origin $(0,0)$. The first point is (u, v) . A circle with (u, v) as center and $a + \delta$ as radius can be shown below.

$$\text{CircleA: } a + \delta = \sqrt{(x-u)^2 + (y-v)^2}$$

To obtain the longest distance $a + \sqrt{2}a$ obtained from Case1, the distance between the first point and the second point must be greater than a . Also, for point (u, v) to be within the red triangle, the distance from point p_1 must be less than or equal to a . As a result, *CircleA* intersects line segment $\overline{p_1p_3}$ at one point and intersects line segment $\overline{p_2p_3}$ at one or two points. These scenarios are discussed as follows. Note, if *CircleA* has no intersection with $\overline{p_1p_3}$ or $\overline{p_2p_3}$, the second point is outside the network area, which does not meet the assumption.

Case2.1: The distance between point (u, v) and point p_2 is less than $a + \delta$ (*CircleA* has only one intersection with $\overline{p_2p_3}$), as shown in Fig. 4a.

Let *CircleA* intersect with $\overline{p_1p_3}$ at point (a, y) , where $(a-u)^2 + (y-v)^2 = (a + \delta)^2$ and $y \leq a$; also let *CircleA* intersect with $\overline{p_2p_3}$ at point (x, a) , where $(x-u)^2 + (a-v)^2 = (a + \delta)^2$ and $x \leq a$. Then,

$$x = \sqrt{(a + \delta)^2 - (a-v)^2} + u \quad (11)$$

$$y = \sqrt{(a + \delta)^2 - (a-u)^2} + v \quad (12)$$

$$db = \sqrt{(a-x)^2 + (a-y)^2} \quad (13)$$

where $a + \delta$ represents the distance from the first point (u, v) to the second point (a, y) , and db represents the distance from the second point to the third point (x, a) . Thus, the total distance *dist* is $dist = (a + \delta) + db$

From Equations (11)-(13), we obtain the following:

- (1) When u becomes larger, and δ, v does not change (that is, the first point deviates to the right), the values of x and y become larger, and db becomes smaller, resulting in a smaller *dist*.
- (2) When v becomes larger, and δ, u does not change (that is, the first point moves upward), the values of x and y become larger, and db becomes smaller, resulting in a smaller *dist*.
- (3) When u, v becomes larger, and δ does not change (that is, the first point moves to the upper right), the values of x and y become larger, and db becomes smaller, resulting in a smaller *dist*.
- (4) When δ becomes larger, and u, v does not change (that is, the distance between the first point and the second point becomes larger), the values of x and y become larger, and db becomes smaller, resulting in a smaller *dist*.

This means that u, v, δ is the smallest, and *dist* is the largest. As a result, when $u = 0, v = 0$, and $\delta = 0$, the *dist* is the largest. The maximum distance of three points is thus $a + \sqrt{2}a$.

Case2.2: The distance between point (u, v) and point p_2 is no less than $a + \delta$ (*CircleA* has two intersection with $\overline{p_2p_3}$).

Let *CircleA* intersect with $\overline{p_1p_3}$ at point (a, y) , where $(a-u)^2 + (y-v)^2 = (a + \delta)^2$ and $y \leq a$. Also, *CircleA* intersects with $\overline{p_2p_3}$ at points (x_1, a) and (x_2, a) , and $x_1 \leq a, x_2 \leq a$. Fig. 4b shows this situation. In this case, the second point (a, y) to the third point $p_2(0, a)$ is the longest distance of the second segment. Then, we have

$$y = \sqrt{(a + \delta)^2 - (a-u)^2} + v \quad (14)$$

$$db = \sqrt{a^2 + (a-y)^2} \quad (15)$$

$$dist = (a + \delta) + db. \quad (16)$$

As in the analysis in Case 2.1, the maximum value of the total distance $dist$ is $a + \sqrt{2}a$.

Finally, by adding twice the longest distance from any point in the area to the central BS, we obtain

$$L_{path}^3 \leq a + \sqrt{2}a + 2 \frac{\sqrt{2}}{2} a = a + 2\sqrt{2}a. \quad (17)$$

Similarly, this value is less than the value of $(\sqrt{3}+1)a + 2\sqrt{2}a$ obtained by Equation (10).

Theorem 3. In a square network area with side length a , four points, x, y, z , and w , are arbitrarily selected. Let L_{path}^4 be the path length starting from the BS and returning to the BS after passing points x, y, z , and w in the nearest point first order. If the BS is located in the center of the network area, then $L_{path}^4 \leq a + 3\sqrt{2}a$.

Proof of Theorem 3. According to Theorem 2, in a square area with side length a , the longest path length is $a + \sqrt{2}a$ in case any three points are visited in the nearest point first order. Since the longest distance between any two points in the area is $\sqrt{2}a$, the longest path length will be less than or equal to $a + 2\sqrt{2}a$ if any four points are visited in the nearest point first order.

Lastly, the doubles of the longest distances from any point in the area to the central BS are added, yielding

$$L_{path}^4 \leq a + 2\sqrt{2}a + 2 \frac{\sqrt{2}}{2} a = a + 3\sqrt{2}a. \quad (18)$$

Note that this value is less than the value of $3a + 2\sqrt{2}a$ obtained by Equation (10).

To further verify equation (10), N_p points ($N_p = 2 \sim 100$) are randomly generated in a $500 \text{ m} \times 500 \text{ m}$ and $1000 \text{ m} \times 1000 \text{ m}$ area. The full path length is defined as the distance starting from the BS using the nearest point first algorithm to visit N_p points and then return to the BS, which is $L_{path}^{N_p}$.

Fig. 5 shows the curves of Equation (9) (with the curve denoted as “Jaillet”) and Equation (10) (with the curve denoted as “Proposed”). The curve denoted as Simulation (LL) / Simulation (C) represents the maximum simulation values obtained from one million executions when the BS is located at the lower left corner / center of the network area. Also shown is the curve denoted Tomar, which is the result of one million simulations based on a formula in [6], with sensor size being 2,000 under $500 \text{ m} \times 500 \text{ m}$ or $1000 \text{ m} \times 1000 \text{ m}$ environments.

Fig. 5 reveals that Equation (10) can obtain a tighter upper bound of the NJF path length than the paths depicted in Equation (9) and derived using the formula proposed in [6] for both network size scenarios. Accordingly, Equation (10) proposed in this paper could provide a good reference for the path length estimation obtained by NJF scheduling.

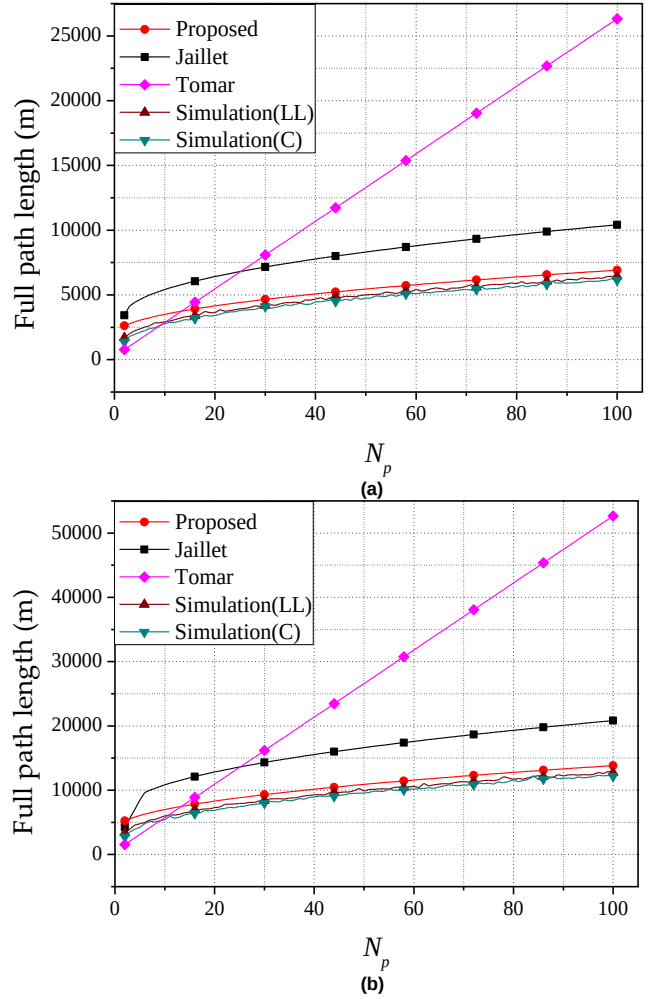


Figure 5. Relationship between the number of nodes (N_p) and full path length in different map sizes. (a) $500 \text{ m} \times 500 \text{ m}$, and (b) $1000 \text{ m} \times 1000 \text{ m}$.

C. THE PROPOSED PROACTIVE CHARGING STRATEGIES

Assume that the WCV is capable of completing a maximum of $N_{maxTask}$ charging tasks on a charging mission, which starts when the WCV leaves the BS, completes all the charging tasks, and returns to the BS. This assumption can be fulfilled if the following two conditions hold: (a) The threshold of the remaining work time for a sensor to issue a charging request is set according to Equation (7), and (b) the frequency of charging requests agrees with Equation (5). In case the number of tasks in the waiting queue equals N_{on_demand} , the number of proactive charging tasks that can be performed is

$$N_{proactive} = N_{maxTask} - N_{on_demand}. \quad (19)$$

An overview of the proposed scheduling algorithm is presented below.

Algorithm 1. Overview of the proposed scheduling algorithm

Input: $clist$ (the list of charging requests), $N_{maxTask}$

Output: $rlist$ (the charging scheduling result).

- 1: Set $N_{proactive}$ to be $N_{maxTask} - N_{clist}$.
- 2: **if** $N_{proactive}$ is greater than zero **then**
- 3: Add $N_{proactive}$ proactive charging tasks to $clist$.
- 4: Set $rlist$ to be the scheduling result of $clist$ by using the NJF / TADP / EDF scheduling algorithm.

5: end

However, selecting the target sensors for proactive charging is an issue that deserves careful consideration. In this section, we present a bottleneck prediction and removal (BP&R) mechanism that predicts and removes potential occurrences of bottlenecks, such as too many potential charging requests within a given period that could overwhelm the charging capacity of a WCV.

1) DEADLINE ESTIMATION

To determine the bottleneck, each sensor needs to estimate its deadline for issuing a charge request properly. However, the power consumption rate of a sensor may be steady or changing depending on the occurrences of incidents. This work proposes a deadline estimation mechanism that imposes a minimum extra communication burden on the network. First of all, each sensor x sends a startup packet with an estimated deadline $deadline_x$ to the BS when it is activated. The BS stores the received $deadline_x$ for any given sensor x and stores the information in $STable$. Second, on each charging mission, the WCV collects the latest estimated $deadline_x$ of each sensor x that has been charged and reports to the BS when returning to the base station. The BS can then update the $STable$ with the newer information.

Finally, each sensor x also maintains the latest deadline information that it reported to the BS (by sending a deadline update packet or collecting it from the WCV during the charging process) as $latest_deadline_x$. When sensor x finds that its power consumption rate $consumption_rate_x$ has changed, it calculates the $new_deadline_x$ accordingly. The gap between the $new_deadline_x$ and $latest_deadline_x$ of sensor x is defined below:

$$diff_x = |new_deadline_x - latest_deadline_x|.$$

If $diff_x$ is greater than the pre-defined threshold $T_{deadlineThreshold}$, sensor x updates its $latest_deadline_x$ and then sends a packet to BS to update the $STable$. Note, the WCV will carry the latest $deadline_x$ back to the BS when sensor x is fully charged. The detailed algorithm is presented as Algorithm 2.

Algorithm 2. The charging deadline information maintenance mechanism performed in each sensor x

Input: $latest_deadline_x, consumption_rate_x, T_{req}$.
Output: none.

- 1: Set $new_deadline_x = current_time + residual / consumption_rate_x - T_{req}$.
- 2: Set $diff_x = |new_deadline_x - latest_deadline_x|$.
- 3: **if** ($diff_x \geq T_{deadlineThreshold}$ and sensor x is not being charged) **then**
- 4: Set $latest_deadline_x = new_deadline_x$.
- 5: **if** (sensor x is not being charged) **then**
 Send a deadline update packet with $latest_deadline_x$ to the BS.
- 6: **end**
- 7: **end**

2) BOTTLENECK PREDICTION AND REMOVAL

It is possible that the WCV will fail to complete some of the charging tasks of a mission in time if there are too many requests over a short period. To be more specific, according to Equation (8), the aforementioned scenario happens if there are more than $N_{maxTask}$ requests initiated within the period of

a WCV's charging mission, $T_{job}(N_{maxTask})$. The bottleneck prediction and removal (BP&R) mechanism is proposed to avoid this potential scenario. The BP&R mechanism works by checking if there will be more than $N_{maxTask}$ charging requests in a certain $T_{job}(N_{maxTask})$ period of time in the future. If one is found, BP&R selects the sensor nodes that potentially issue charging requests during that period as the proactive charging seeks to alleviate the possible bottleneck for later WCV charging missions.

We illustrate the idea of the BP&R mechanism by presenting a more concrete example below. Fig. 6 shows the occurrences of 15 charging requests at various points on time. Numbers 1–15 represent the sensor IDs that issue requests. Assume that $N_{maxTask}$ equals 10, indicating that each WCV charging mission, which takes $T_{job}(N_{maxTask})$ seconds, is capable of completing 10 charging tasks. The BP&R works similar to a sliding window, with $T_{job}(N_{maxTask})$ being the window size. The BP&R keeps sliding the window forward, stopping at every time point with a charging request, and counts the number of requests within that window. For example, Window 1 (in Fig. 6) contains 8 requests (with IDs numbered from 1 to 8), which is acceptable, as it is less than $N_{maxTask}$ ($= 10$). However, Window 4 contains 11 requests (with IDs numbered from 4 to 14), which is more than one WCV mission can handle. Accordingly, the BP&R mechanism will randomly choose a sensor within that period (e.g., sensor 14) as the target(s) for proactive charging to eliminate a possible bottleneck scenario. The charging requirement of sensor 14 can then shift backwards to the next charging cycle and maintain the number of charging requests in that period as less than or equal to $N_{maxTask}$ ($= 10$).

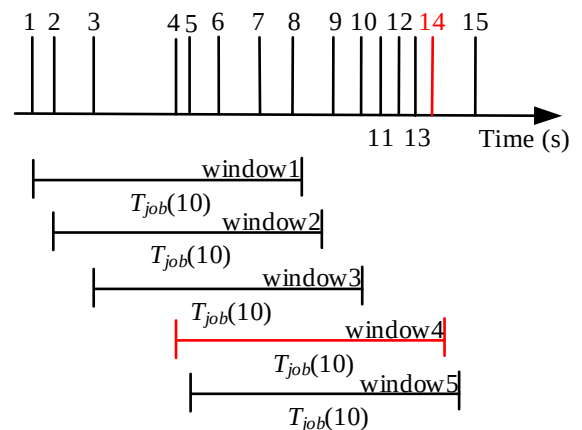


Figure 6. An example illustrating the Bottleneck Prediction and Removal Mechanism.

If the power consumption rate of the sensors is not constant, the BP&R mechanism works somewhat differently. As stated earlier, a sensor will send a packet to the BS to update its deadline information when the estimated deadline variation exceeds a certain threshold, $T_{deadlineThreshold}$. Therefore, the possible time that a sensor may issue a charging request is now specified by the estimated time plus

/ minus $T_{deadlineThreshold}$. As an example, the dotted line in Fig. 7 indicates the range on time that sensor 1 can possibly issue its charging request. The BP&R then counts the number of ranges covered in every sliding window. For example, Window 1 (in Fig. 7) contains not only the first eight requests (with IDs numbered from 1 to 8), but also request 9 as part of its left bound located within Window 1. However, Window 4 contains 12 potential charging requests (with IDs numbered from 4 to 15), which is greater than $N_{maxTask}$. Therefore, the BP&R mechanism randomly chooses two sensors within Window 4 as the target(s) for proactive charging to eliminate this potential bottleneck.

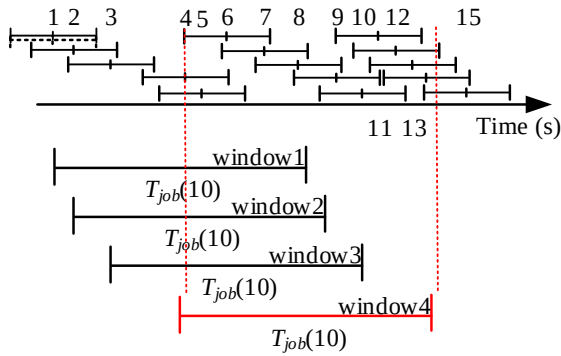


Figure 7. An example illustrating the Bottleneck Prediction for a set of time ranges

The above bottleneck problem can be modeled by using interval graphs. $G = (V, E)$ is an interval graph [34] provided that one can assign an interval I_u to each vertex u such that $I_u \cap I_v$ is nonempty precisely when $(u, v) \in E$. For example, in Fig. 8, the graph is an interval graph because there is a set of intervals representing a vertex set. A clique [35] in the graph represents a set of pair-wise adjacent vertices.

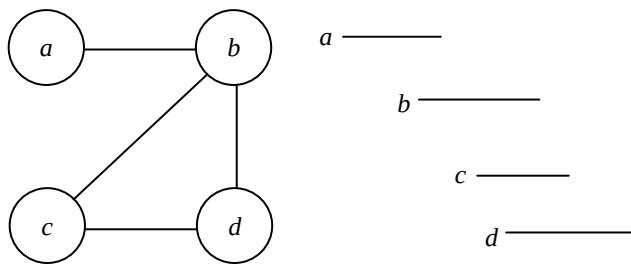


Figure 8. An interval graph.

The aforementioned bottleneck problem can be modeled by using the above interval graph G such that each vertex represents a charging request, which is represented by an interval $[t - T_{deadlineThreshold} - T_{job}(N_{maxTask})/2, t + T_{deadlineThreshold} + T_{job}(N_{maxTask})/2]$ in the interval model, where t is the deadline of the charging request. The bottleneck problem now involves removing the fewest number of requests (intervals) such that the resulting corresponding interval graph does not present a clique with a size greater than or equal to $N_{maxTask}$.

The detailed algorithm is presented in Algorithm 3.

Algorithm 3. The proposed BP&R mechanism

Input: $clist$ (The list of charging requests), $N_{maxTask}$, $STable$.
Output: $cplist$ (The list of charging requests with proactive charging tasks).

- 1: Set $N_{proactive}$ to be $N_{maxTask} - N_{clist}$.
- 2: **if** $N_{proactive} \geq 0$ and $N_{Slist} \geq 0$ **then**
- 3: Set $Slist$ to be the ascendingly sorted sensor list according to the remaining work times stored in the $STable$.
- 4: Set $cplist = clist$.
- 5: **for** each sensor x in $Slist$ **do**
- 6: $st = T_{deadline,x}$.
- 7: $et = st + T_{job}(N_{maxTask})$.
- 8: $Interval_x = [st, et]$.
- 9: $BottleList = NULL$.
- 10: **for** each sensor y in $Slist$ **do**
- 11: $Interval_y = [T_{deadline,y} - T_{deadlineThreshold}, T_{deadline,y} + T_{deadlineThreshold}]$
- 12: **if** $(Interval_x \cap Interval_y \neq \{\})$ **then** Add y into $BottleList$.
- 13: **end**
- 14: **if** $(N_{BottleList} > N_{maxTask})$ **then**
- 15: $RN = \min(N_{proactive}, N_{BottleList} - N_{maxTask})$.
- 16: Set $Addlist$ to be the list of RN sensors randomly selected from $BottleList$.
- 17: Add the sensors in $Addlist$ to $cplist$.
- 18: Remove the sensors in $Addlist$ from $Slist$.
- 19: **end**
- 20: **if** $(N_{clist} \geq N_{maxTask})$ **then return**
- 21: Set $N_{proactive}$ to be $N_{maxTask} - N_{clist}$.
- 22: **end**
- 23: **end**

The time complexity of Algorithm 3 is analyzed as follows: Step 1 and Step 2 can be executed in $O(1)$ time. Step 3 can be accomplished in $O(n \log n)$ time by applying an optimal sorting algorithm, where n is the number of charging requests. Step 4 copies n entries/requests from a list to another, which can be done in $O(n)$ time.

The outer for-loop from Step 6 to 21 needs to be executed $O(n)$ times. The steps for the body of the outer for-loop are discussed below. Steps 6–9 can be trivially computed in $O(1)$ time. For Step 10, Step 11, Step 12, and Step 13 to form an inner for-loop takes $O(n)$ time. Steps 14 – 22 take $O(n)$ time in total to execute the outer for-loop. Finally, Step 23 can be completed in $O(1)$ time. Overall, Algorithm 3 can be executed in $O(n^2)$ time. Note that the time complexity of Algorithm 3 can be further reduced to $O(n \log n)$ time by applying a sophisticated data structure called *interval trees* [36] to determine overlapping intervals.

V. SIMULATION RESULTS

The simulation program was coded using Visual Studio C# and executed on a personal computer running Microsoft Windows 10 with an Intel i7-8700 CPU. The simulation environment is a 500 m × 500 m or 1000 m × 1000 m square area with up to 3000 sensors evenly and randomly deployed. The BS is located in the lower left corner at coordinates (0, 0). The WCV is equipped with a limited capacity power supply for its movement and charging tasks. We summarize the other parameters used for the simulation in Table II.

TABLE II
SIMULATION PARAMETER SETTINGS

Parameters	Values
Simulation time	500000 s

Number_of_Sensors	2000
Energy_Capacity_of_Sensor	100 J
Baseline energy consuming rate of sensor	10^{-3} J/s
Lifetime_of_Sensor	100000 s
Number of WCVs	1
Energy capacity of WCVs	200000 J
Energy consumption of moving WCVs	10 J/m
Traveling speed of WCVs	5 m/s
Charging rate	5 J/s

As shown in Table III, we have organized and compiled a table listing simulation parameters used in other notable works, with highlighted numbers (in bold) being ones that are the same as (or similar to) our settings. In particular, our parameters, such as network size, WCV travelling speed and charging rate, are mainly referring to those used by Xie et al. [1]. However, we place more sensor nodes in the network and provide less sensor's battery power so that there are more frequent charging needs. In addition, the WCV batter energy is set higher as it has more ground to cover.

TABLE III
PARAMETER SETTINGS USED IN OTHER METHODS

	[1]	[6]	[7]	[8]	[15]
Battery full	10.8 kJ	10.8 kJ	10 J	100 kJ	500 J
Network size(m ²)	1000×1000	100×100	200×200	100×100	100×100
Number of node	50, 100	400~1000	100	100	20~100
WCV traveling speed (m/s)	5	1~7	5	4	3
WCV energy	–	4×10 ⁶ J	5000J	–	50000J
WCV moving consumption	–	600 J/m	4 J/m	–	10 J/m
Charging speed	5 J/s	5 J/s	200 mJ/s	–	5 J/s

NJF is the primary scheduling algorithm used in our simulations. EDF [5] and TADP [8] are also included in some simulations for comparison purposes. Network lifetime, defined as the time when the first sensor dies, is the major performance metric used in this paper as it is the most important performance indicator for a WRSN. All data values shown in the tables or charts are the average of 30 simulation results.

An essential parameter, the frequency of charging requirement, p , is defined as the number of sensors that need to be recharged over a sensor's lifetime ($Lifetime_of_Sensor$). For example, as shown in Table II, if there are 2,000 sensors deployed in a network, the time that these sensors need to be recharged is evenly distributed within the $Lifetime_of_Sensor$ (=100000 seconds) period. The frequency of charging requirements, p , is thus 0.02 per second. In later simulations, p is used as the parameter to adjust the loading of the network for the input test files. This scenario works as follows. At the beginning of simulations, the sensors are activated at a probability of $p / Number_of_Sensors$ per second until the required number of sensors, $p \times Lifetime_of_Sensor$, is started. The sensors are fully charged at startup. We assume that all sensors consume their energy at a constant rate [8], [12], which

equals $Energy_Capacity_of_Sensor / Lifetime_of_Sensor$. Accordingly, each sensor will drain its battery and die in $Lifetime_of_Sensor$ seconds after its activation if there is no appropriate recharging action taken.

Note, to reduce the number of variables, the following simulations first assume a fixed sensor power consumption rate scenario. This assumption is lifted and the performance of the proposed algorithm under a variable sensor power consumption rate is evaluated in Section V.D.

A. INPUT TEST FILES FOR SIMULATION

According to Equation (5), the upper bound of the charging request frequency that NJF can safely handle depends on the appropriate values of $N_{maxTask}$ and T_{req} . We set $N_{maxTask}$ to 30, which is a conservative yet reasonable value (according to Fig. 5b and Table II) for a 1000 m × 1000 m network environment. To be more specific, in this case, the WCV moves about 9000 meters and charges 30 sensors, which consumes approximately 60% of its battery power (12000J). Once the value of $N_{maxTask}$ is determined, we can derive the lower bounds of T_{req} as approximately 4620 s and 2900 s from Equation (7) when the network sizes are 1000 m × 1000 m and 500 m × 500 m, respectively. The upper bounds of the charging request frequency that NJF can safely handle thus corresponds to 0.012 (1000 m × 1000 m) and 0.0196 (500 m × 500 m). Without any proactive measure, the frequency of the charging requests (F_{req}) will be the same as the frequency of charging requirements (p).

However, the proactive charging strategies do affect the charging order or the remaining power of the nodes in the later recharge cycle. It is thus not simple to establish a test file with the frequency of the charging requirements (p) strictly controlled using an exact value. However, it is possible to set this value within the range of the p value. In this paper, we calculate the frequency distribution using a sliding window, which we also adopt for the BP&R mechanism in Section IV.C. We set the window size to 10 and calculate the frequencies by averaging the inverses of the time gaps between every 10 consecutive charging requirements. Taking Fig. 9 as an example, there are 15 potential charging requests (i.e., charging requirements) between 10 and 78 seconds. When the window size equals 10, the frequencies denoted as f_1 - f_6 can be derived accordingly. The average of these six frequencies is 0.186167, with the maximum and minimum being 0.2 and 0.167. If 0.02 is used as the statistical interval, the number of occurrences of frequency ranges [0.16, 0.18), [0.18, 0.2), and [0.2, 0.22) will all be twice.

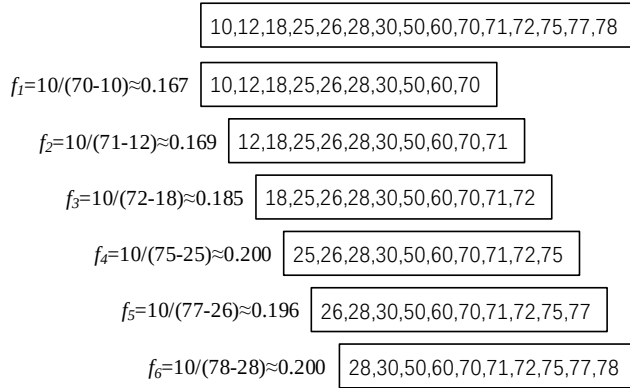


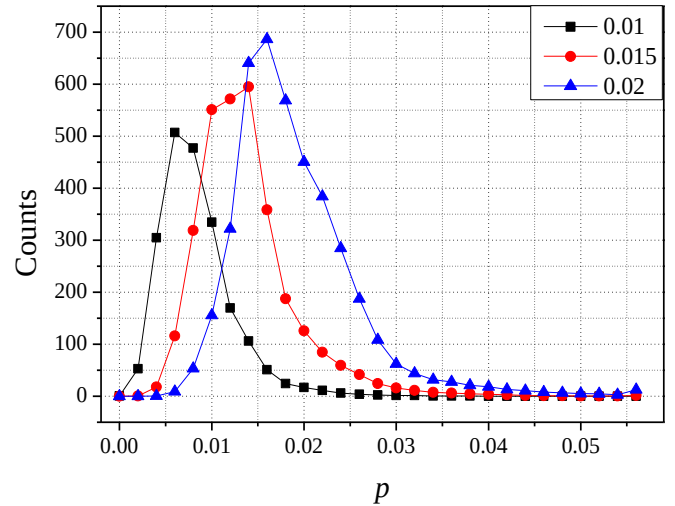
Figure 9. An example using sliding windows to calculate the frequencies of charging requests in the test files.

Table IV lists the average, minimum, and maximum values of the sampled input test files used in this paper. Fig. 10 presents the occurrences of frequency range distributions with a statistical interval that equals 0.005. The data presented in both Table IV and Fig. 10 are derived using the NJF scheduling algorithm. As noted in both cases, the setting of the p value in our test files only represents an approximate average of the distribution of charging needs. In actual simulation scenarios, the frequency of charging needs (and thus charging requests) can be less or significantly more than the p value. For example, in a 1000 m $\times$ 1000 m case with $p=0.015$, the maximum p value can be more than six times of the average. In addition, the timing for a sensor to issue requests in consecutive charging cycles may be affected by the charging scheduling methods adopted. Therefore, the results presented in Table IV and Fig. 10 may be slightly different under EDF or TADP scheduling.

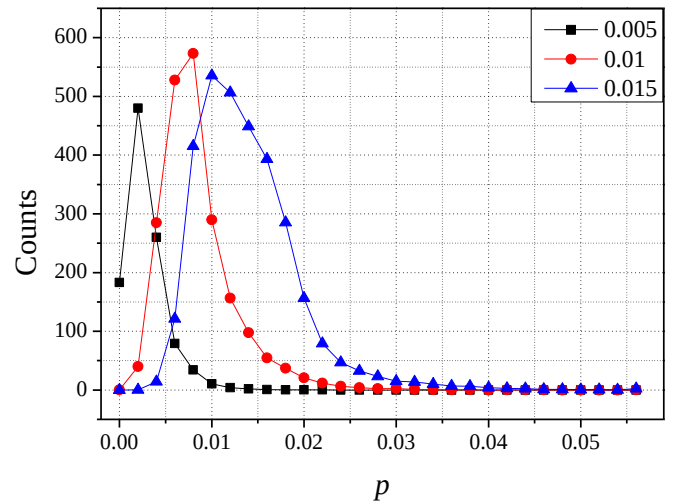
TABLE IV

THE AVERAGE, MAXIMUM, AND MINIMUM VALUES OF THE INPUT TEST FILES WITH VARIOUS p VALUES.

Map size	p	Average	Max	Min
500 m $\times$ 500 m	0.010	0.010263	0.061728	0.003428
	0.015	0.015366	0.095238	0.005672
	0.020	0.020353	0.104167	0.006812
1000 m $\times$ 1000 m	0.005	0.005202	0.031546	0.002057
	0.010	0.010372	0.063291	0.003414
	0.015	0.015328	0.111111	0.005300



(a)



(b)

Figure 10. The frequency distributions of some test files. (a) map size: 500 m $\times$ 500 m; (b) map size: 1000 m $\times$ 1000 m.

B. PERFORMANCE IMPACT OF T_{req}

The values of T_{req} , as calculated in the previous section, are 2900 s and 4620 s with a $N_{maxTask}$ equal to 30 and a frequency of charging requirement (p) corresponding to 0.0196 and 0.012 for network sizes of 500 m $\times$ 500 m and 1000 m $\times$ 1000 m, respectively. To the best of our knowledge, there is no paper discussing how to set the T_{req} value. Some papers set sensors to issue charging requests in their simulations when the sensors have less than 15%~40% power remaining [8] - [12]. Assume that the sensor consumes power at a fixed rate, with the sensor lifetime set to 100,000 seconds (as shown in Table II); 20% of the remaining power is then equivalent to the remaining working time being 20,000 seconds, which seems to be too high. In addition, some studies [6], [7] use the average distance of any two points in the network to estimate the moving distance of the WCV, which may result in greatly misaligned outcomes. As an example, with the sensor size set to 2,000, the average distance $D_{average}$ of any two points under

a $500\text{m} \times 500\text{m}$ or $1000\text{m} \times 1000\text{m}$ environment may be 260.7 m or 521.4 m, respectively, when calculated with the method in [6]. Under our simulation environments, these values correspond to charging thresholds (in residual energy) of 148.5 J and 256.8 J, which are larger than the assumed full battery power of the sensor (100J).

In this section, we evaluate the impacts of T_{req} values on the network lifetime and number of charging requests using the following three methods: NJF, TADP, and EDF. The frequency of charging requirement (p) for the simulation test files is set to slightly less than the upper bound, with p set to 0.015 and 0.010 for the $500\text{m} \times 500\text{m}$ and $1000\text{m} \times 1000\text{m}$ environments.

Fig. 11 presents the simulation results. As shown in Fig. 11a,b, NJF and TADP perform equally well in terms of their network lifetime when T_{req} equals approximately 7~9% of the remaining power. EDF falls far behind in this respect. In addition, as far as NJF is concerned, all charging requests can be completed in time when T_{req} is close to (or beyond) the calculated threshold (2900 s for $500\text{m} \times 500\text{m}$ and 4620 s for $1000\text{m} \times 1000\text{m}$). However, as illustrated in Fig. 11c,d, as T_{req} increases, the sensors tend to issue charging requests sooner, resulting in more charging tasks and power consumption for the WCV.

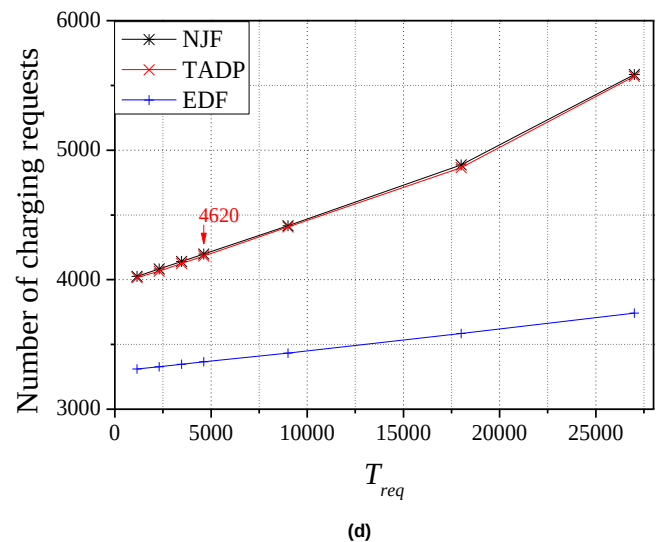
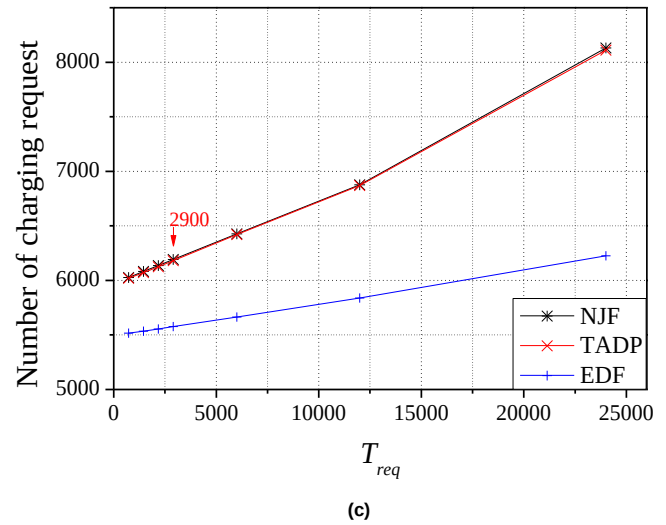
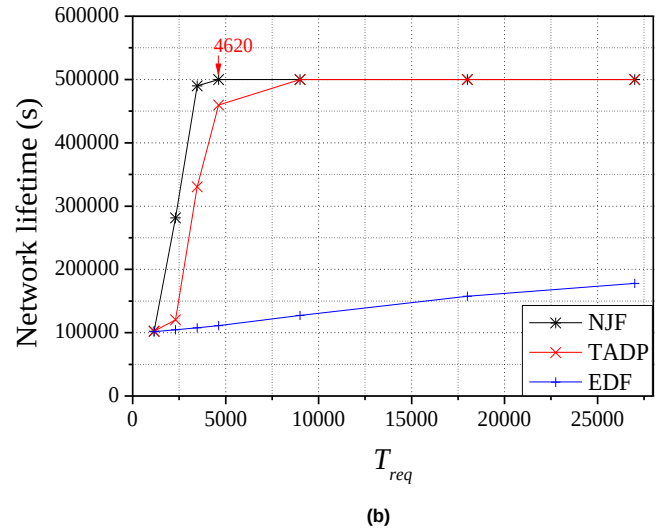
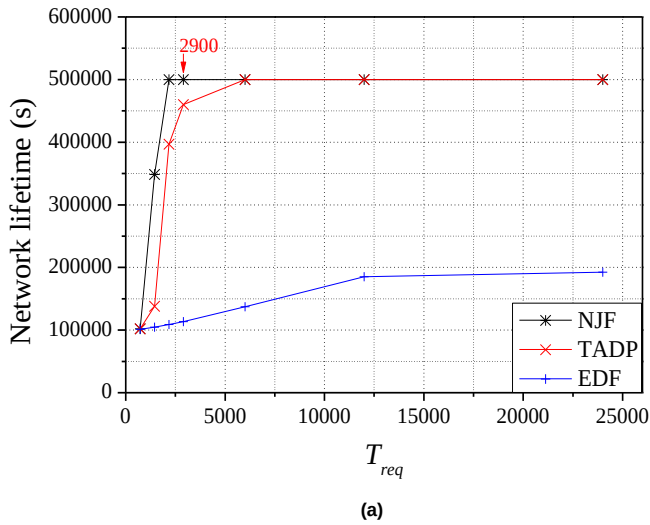


Figure 11. The impact of setting different values of T_{req} on the network lifetime and number of charging requests under different map sizes. (a), (c) map size: $500\text{m} \times 500\text{m}$; (b), (d) map size: $1000\text{m} \times 1000\text{m}$.

C. PERFORMANCE COMPARISONS OF EDF, TADP, AND NJF WITH / WITHOUT THE BP&R MECHANISM

In this section, we put together EDF, TADP, and NJF to assess how they perform with or without the use of a BP&R proactive mechanism. To simulate scenarios under normal to extreme loading conditions, p is set between 0.013 and 0.024 or 0.005 and 0.016 for the 500 m $\times$ 500 m or 1000 m $\times$ 1000 m environment, respectively.

The T_{req} settings (2900 s for 500 m $\times$ 500 m and 4620 s for 1000 m $\times$ 1000 m) in the simulations are based on the moving distance of WCV using Equation (10), which relies on the assumption that path planning is done using the NJF algorithm. Since EDF scheduling does not take into consideration the distance factor, the moving distance for completing a charging mission would be greater than that estimated in Equation (10). The same also holds for TADP scheduling, which is not entirely depending on the distance factor. Accordingly, the aforementioned T_{req} settings are not in favor of the EDF and TADP scheduling methods. In other words, both require larger T_{req} values that are set in simulations, as shown in Fig. 11. However, this is also an opportunity to investigate the benefits of our proposed BP&R proactive selection mechanism. Fig. 12 presents the results in terms of network lifetime. Note that NJF/P, TADP/P, and EDF/P indicate that NJF, TADP, and EDF scheduling are operating with the BP&R proactive mechanism.

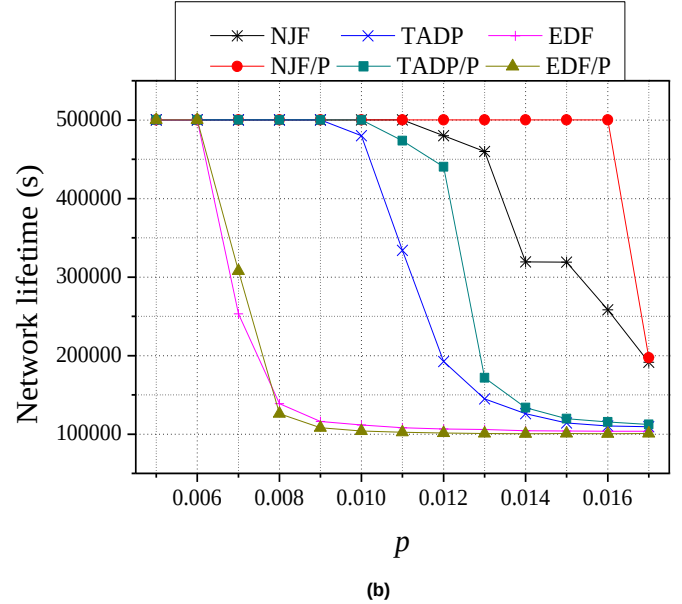
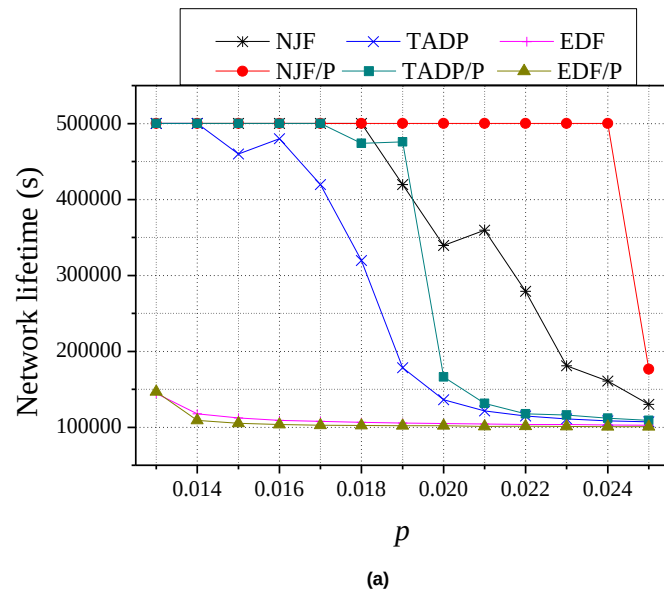


Figure 12. Simulation results for the network lifetime under different map sizes. (a) map size: 500 m $\times$ 500 m; (b) map size: 1000 m $\times$ 1000 m.

As expected, NJF performs better than the other two methods in both network environments. However, as the frequency of charging needs (p value) increases, all three methods seem to be unable to handle the loadings well. With the BP&R proactive mechanism, both NJF and TADP can extend the network lifetime under heavier loading conditions. This improvement is far more significant for NJF. On the other hand, BP&R does not have any effect on EDF scheduling. To explore the effect of proactive charging further, we list the number of on-demand (N_{on_demand}) and proactive ($N_{proactive}$) charging tasks that BP&R works with under the three scheduling approaches in both network size scenarios, with varying p values provided in Table V and VI. Note, a WCV's charging mission usually contains N_{on_demand} on-demand plus $N_{proactive}$ charging tasks.

Table V and VI show that the number of on-demand charging requests (N_{on_demand}) increases as the charging needs increase. For EDF, the requests (N_{on_demand}) pile up so quickly that they soon overwhelm the charging capacity of the WCV. This is understandable since EDF does not consider the distance factor, which makes the WCV spend more time and energy on moving long distances without being able to complete its charging tasks in time. This results in queuing more subsequent charging requests, which leaves no room for BP&R engagement. TADP scheduling, which takes into account the time and distance factors equally, performs better than EDF. We can see that BP&R intervenes, though not in a significant way, and helps remove some potential bottlenecks to extend the network lifetime (when p values are around 0.018 in the 500 m $\times$ 500 m case or 0.012 in the 1000 m $\times$ 1000 m case). However, when the loading scales up to a higher level (e.g., $p \geq 0.021$ for 500 m $\times$ 500 m or $p \geq 0.013$ for 1000 m $\times$ 1000 m), the on-demand requests soon exceed the WCV's capability.

TABLE V

THE AVERAGE NUMBER OF ON-DEMAND TASKS AND PROACTIVE TASKS
WITH VARYING p (MAP SIZE: 500 M $\times$ 500 M).

p		0.018	0.020	0.020	0.022	0.024	0.025
NJF	N_{on_demand}	8.35	12.19	19.07	20.89	22.83	26.34
/P	$N_{proactive}$	0.86	3.32	10.92	9.10	7.19	4.94
TADP	N_{on_demand}	10.50	18.63	28.82	36.85	53.54	65.59
/P	$N_{proactive}$	0.41	1.69	2.60	0.77	0.19	0.22
EDF	N_{on_demand}	259.2	338.1	595.5	770.4	948.6	1028.2
/P	$N_{proactive}$	0.38	0.378	0.37	0.38	0.38	0.38

TABLE VI

THE AVERAGE NUMBER OF ON-DEMAND TASKS AND PROACTIVE TASKS
WITH VARYING p (MAP SIZE: 1000 M $\times$ 1000 M).

p		0.008	0.010	0.012	0.014	0.016	0.017
NJF	N_{on_demand}	5.88	9.18	17.59	19.11	20.45	24.57
/P	$N_{proactive}$	0.16	1.26	12.38	10.87	9.53	6.55
TADP	N_{on_demand}	6.54	12.07	26.56	34.32	51.90	63.59
/P	$N_{proactive}$	0.09	0.65	3.59	1.51	0.28	0.28
EDF	N_{on_demand}	58.08	228.7	402.3	577.5	754.1	842.2
/P	$N_{proactive}$	0.20	0.33	0.33	0.33	0.33	0.33

Finally, the NJF scheduling works as we estimated in previous sections. The number of proactive tasks ($N_{proactive}$) increases gradually according to the loadings, indicating that the BP&R mechanism has identified and removed more future bottlenecks by selecting sensors for proactive recharging. When the loading becomes heavier ($p \geq 0.021$ for 500 m $\times$ 500 m or $p \geq 0.013$ for 1000 m $\times$ 1000 m), the number of proactive tasks drops as the number of on-demand requests (N_{on_demand}) approaches $N_{maxTask}$, which is 30 in our settings. In addition, the sum of N_{on_demand} and $N_{proactive}$ approximately equals 30.

Fig.13 shows the relationship between the frequency of charging requests and charging needs. This can allow us to visualize the effects of the BP&R mechanism from different perspectives.

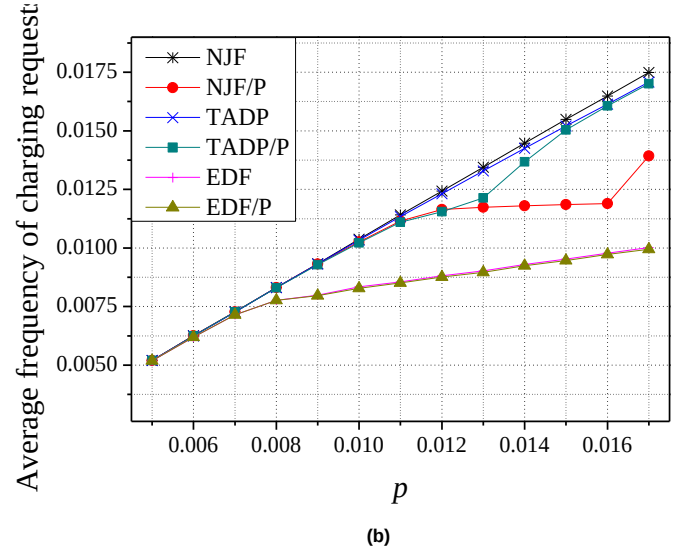
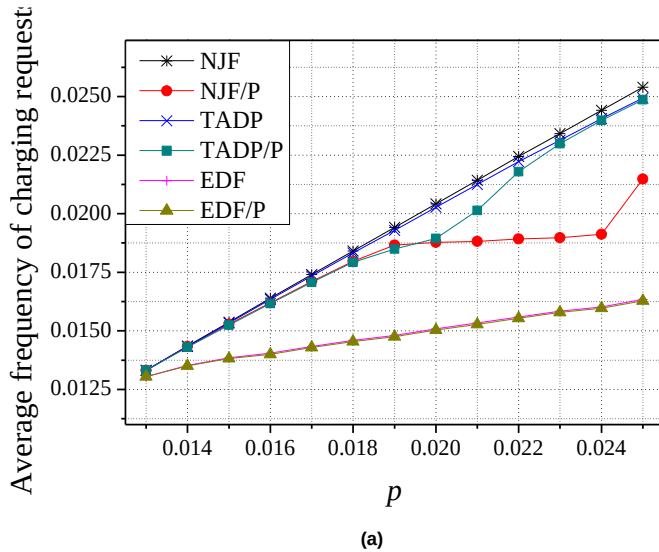


Figure 13. Simulation results for the average frequency of charging requests issued during the simulation processes under different map sizes. (a) map size: 500 m $\times$ 500 m; (b) map size: 1000 m $\times$ 1000 m.

As stated earlier, without external interference, the actual charging requests issued are the same as the charging needs. The straight line denoted by NJF demonstrates this fact. However, with the BP&R proactive mechanism, the average frequency of charging requests for NJF and TADP may be reduced by up to 21.6% / 6.6% or 27.9% / 8.7% in the 500 m $\times$ 500 m or 1000 m $\times$ 1000 m case, respectively. These reductions in charging requests correspond to increases in the network lifetime of NJF and TADP by up to 210.6% / 166.8% or 93.6% / 128.8% (see Fig.12) in the two network environments.

Another important metric for evaluating the charging strategy is charging efficiency. Charging efficiency can be defined as the ratio of total power transferred (to the sensors) to the total power consumed by the WCV [25]. Fig. 14 shows the charging efficiency of different methods with or without enforcing the BP&R proactive mechanism.

First, we see that the increases in charging efficiency for both NJF and TADP, but not EDF, are approximately proportional to the values of p . As the p value increases, the BS (or WCV) needs to consider more charging requests each time it schedules. As a result, the average moving distance can be better optimized via scheduling methods that consider the distance factor, which then reduce the average consumption of energy for WCV movement between sensors. Secondly, we also observe that the BP&R proactive mechanism improves the original NJF algorithm slightly in terms of its charging efficiency. This is no surprise as the intent of the BP&R proactive mechanism is to use the remaining time and power of the WCV to include additional charging tasks in advance to remove potential bottleneck in the future. Therefore, for the same reason mentioned above, the increased overall charging tasks again benefit a scheduling method that solely considers the distance factor. However, as the value of p increases, indicating that the scheduling pool is filled with more on-

demand requests (and less proactive tasks, as shown in Tables V and VI), the effect of BP&R—and thus the improvement in charging efficiency—reduces. Overall, there is room for improvement in this regard, which will be a major topic of our future research.

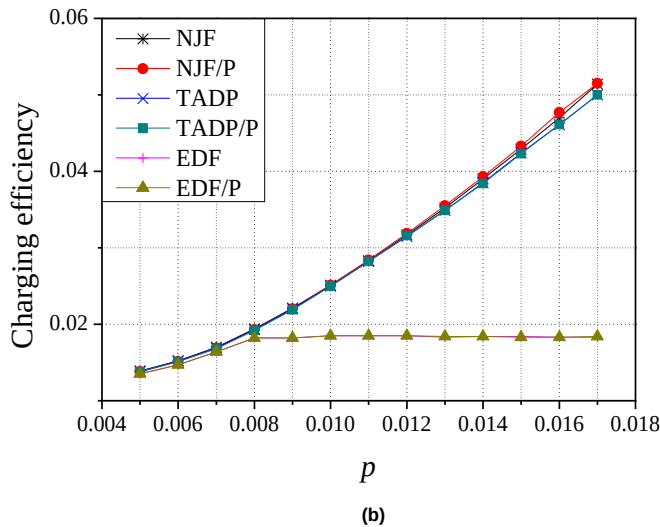
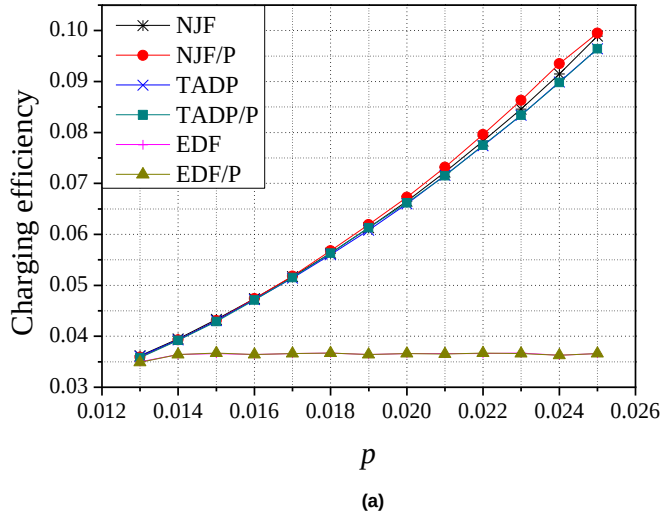


Figure 14. Simulation results for the charging efficiency under different map sizes. (a) map size: 500 m × 500 m, (b) map size: 1000 m × 1000 m.

D. PERFORMANCE OF NJF WITH THE BP&R MECHANISM IN A DYNAMIC POWER CONSUMPTION ENVIRONMENT

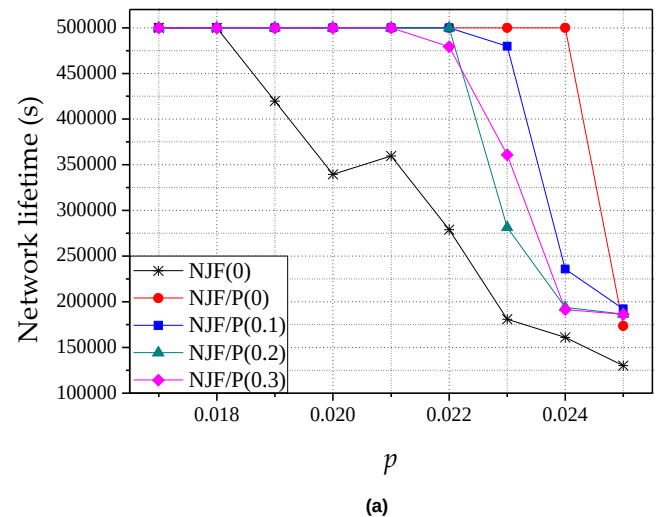
The BP&R mechanism works by predicting and removing potential occurrences of excessive charging requests within the sliding windows of a fixed-size period. For the prediction to be accurate, it needs to be aware of the status of the sensors. In our WRSN model, as described in Section IV.C, the BS is not capable of knowing the exact status of any sensor at any time. It relies solely on the data obtained when the WCV performs its charging task and records those data in the *STable* to estimate the deadline of each sensor. Our assumption (similar to that in [13, 17]) regarding the fixed power

consumption rate of each sensor could help BP&R produce correct decisions. However, the sensors' power consumption rates do change over time in real-life scenarios.

In this section, we simulate changes in the network environment by specifying that each sensor changes its power consumption rate with a probability of P_{rate_change} every 10000 s, with the range of change randomly assigned between -12.5% and +12.5%. Using simulations, we then try to assess the effect of such changes in the power consumption rate on the performance of the BP&R mechanism with varying charging need conditions (p values). In this simulation setting, the p values are set larger to ensure that the BP&R mechanism is triggered.

Fig. 15 presents the simulation results without enforcing the charging deadline information maintenance mechanism. The notation, NJF/P (P_{rate_change}), represents NJF scheduling with the BP&R mechanism enabled and changes in the sensor's power consumption rate with P_{rate_change} probability.

The power consumption rate of the sensors may change the frequency of the charging demand and timing in the occurrences of charging demand bottlenecks. Since NJF considers only the distance factor, changes in the P_{rate_change} value may not have a direct effect on the network lifetime. However, the P_{rate_change} value does prevent BP&R, to a certain degree, from accurately predicting potential bottlenecks in charging demands. This value, as shown in Fig. 15, inevitably results in a negative impact on the network's lifetime. In other words, the network lifetime degrades as the probability of the rate of change (P_{rate_change}) increases. Nevertheless, despite the abrupt variation in the power consumption rate (e.g., when P_{rate_change} equals 0.3), the BP&R proactive mechanism is still capable of improving the network lifetime of NJF scheduling.



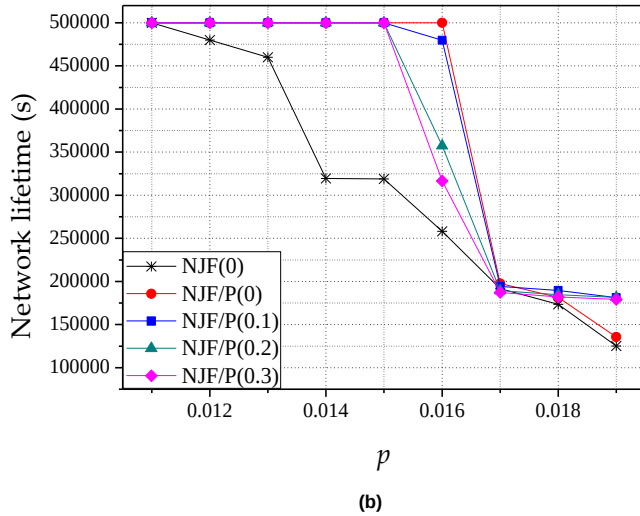


Figure 15. The impact of variation in sensors' power consumption rate on the performance of the BP&R proactive charging mechanism under different map sizes. (a) map size: 500 m x 500 m; (b) map size: 1000 m x 1000 m.

Fig. 16 presents the results with activation of the charging deadline maintenance mechanism (Algorithm 2). The performance in terms of network lifetime tends to be independent of power consumption variations (P_{rate_change} changes from 0.1 to 0.3), which is a significant improvement. This indicates that the proposed charging deadline maintenance scheme can greatly reduce the impact of random changes in the power consumption rate. Note that, in the above simulations, the $T_{deadlineThreshold}$ mentioned in Section IV.C is set to be T_{req} —that is, 2900 s and 4620 s for the 500 m x 500 m and 1000 m x 1000 m maps, respectively.

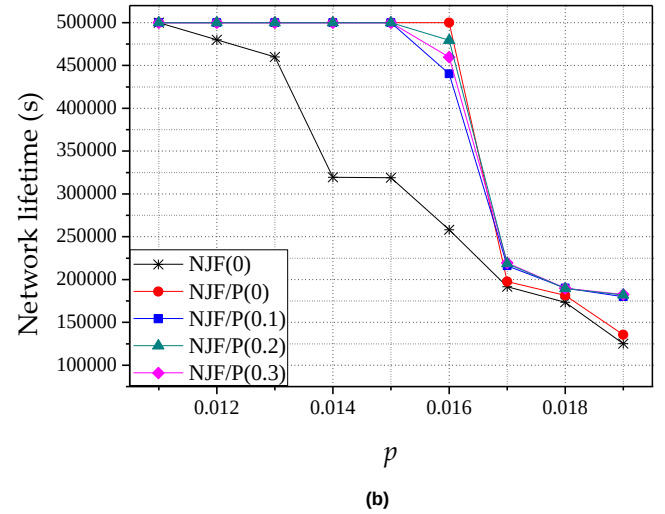
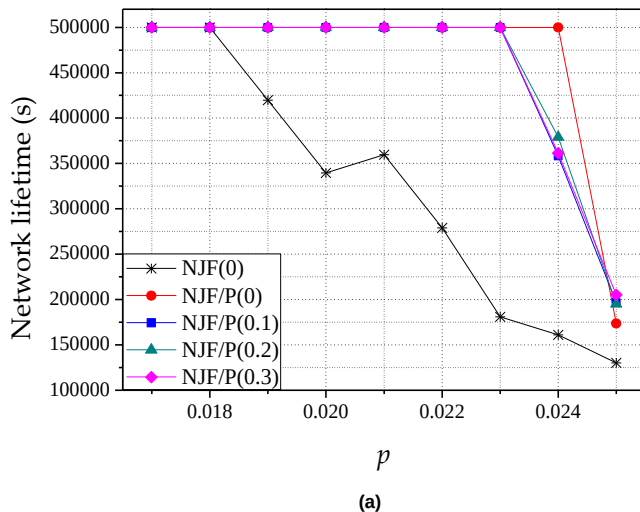


Figure 16. The impact of using the deadline update mechanism on the performance of the BP&R proactive charging strategy under different map sizes. (a) map size: 500 m x 500 m; (b) map size: 1000 m x 1000 m.

Finally, we compare the performance of our proposed method to two recent studies: a Real-Time On-Demand Charging Scheduling Scheme (denoted RCSS hereafter) [15] and a Fuzzy Logic-based On-demand Charging Algorithm (denoted FUZZY hereafter) [6]. RCSS and FUZZY are chosen because they both adopt the energy consumption rate (of sensors) as one of the primary factors for determining charging order, which fits our current simulation context with varying sensor power consumption. More specifically, RCSS combines the energy consumption rate and distance (between the WCV and sensor) with fixed weights to determine the charging order. FUZZY adopts a fuzzy logic mechanism to blend network attributes, such as the sensor's energy consumption rate, residual energy, distance, and node density, to determine the charging schedule of the WCV.

Although both RCSS and FUZZY need mechanisms to estimate or "know" the status of all sensors (including the energy consumption rate), we assume that the BSs of both RCSS and FUZZY are aware of all information of a sensor in real-time in the simulation. However, to ensure a common basis of comparison, we need to tailor both the RCSS and FUZZY algorithms to use the same assumptions as NJF/P. These assumptions include a single available WCV, one sensor being charged at a time, each sensor's battery being charged in full before the WCV proceeds to the next one, and the charging schedule being non-preemptive once determined. The simulation results are presented in Fig. 17.

As shown in Figure 17, the FUZZY method performs better than the RCSS method in terms of the network lifetime. However, regardless of the power consumption variations, the NJF with our proposed BP&R mechanism (NJF/P) performs the best. In particular, when $P_{rate_change}=0$, the network lifetime of NJF/P can be extended by up to 210.56% / 352.50% or 93.58% / 311.61% of the lifetime of FUZZY / RCSS under the 500 m x 500 m or 1000 m x 1000 m network environments, respectively. When $P_{rate_change}=0.3$, the network lifetime of NJF/P can also be extended by up to 167.67% / 281.23% or

107.78% / 266.85% of the lifetime of FUZZY / RCSS under the 500 m × 500 m or 1000 m × 1000 m network environments, respectively.

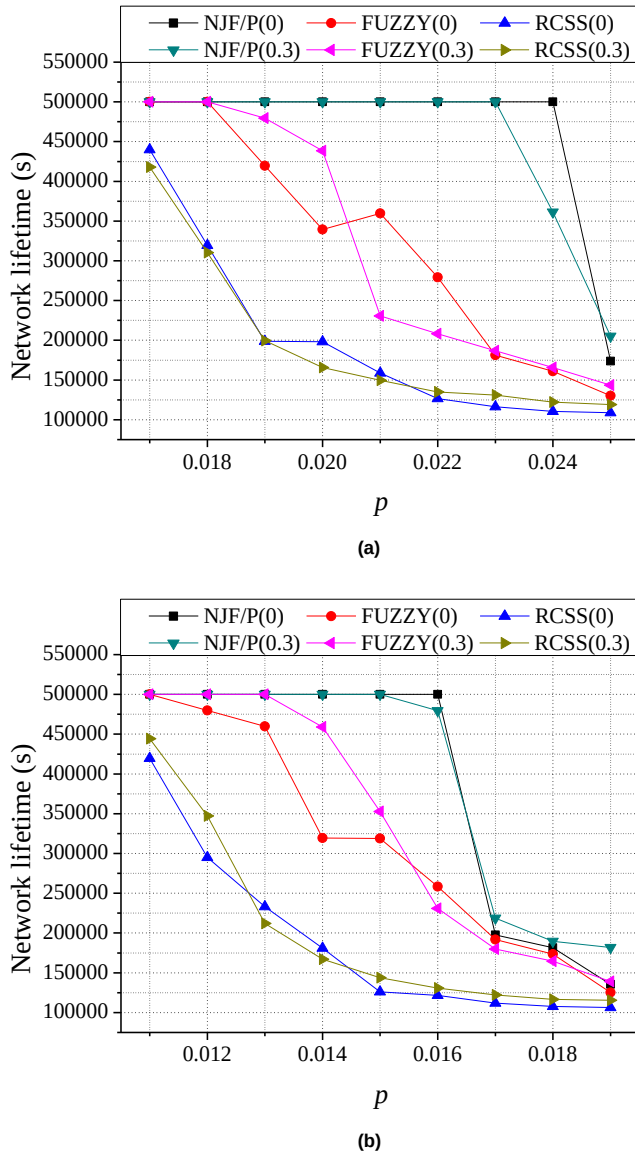


Figure 17. Comparison of our proposed algorithm to RCSS and FUZZY in terms of the network lifetime under different map sizes. (a) map size: 500 m × 500 m; (b) map size: 1000 m × 1000 m.

VI. CONCLUSIONS

Most previous works have attempted to solve the charge scheduling issue by using proactive or on-demand methods. Both scheduling methods have their advantages and disadvantages. Proactive methods may work well if sensors in WRSNs operate with stable power consumption rate; however, it may not adapt well to the dynamic change of network operations. On the other hand, the on-demand methods do determine the distribution of charging demands before acting upon the requests. However, if there are too many charging requests in a short period, a potential bottleneck may make the WCV fail to meet all its charging requests in time.

In this work, we propose a bottleneck prediction and removal mechanism (BP&R) that seamlessly integrates the two completely different scheduling strategies mentioned above. Most previous on-demand scheduling methods consider time (such as deadline or threshold for a sensor to get recharge, sensor's residual, and sensor's power consuming rate, are all related to time factor), distance (the travel distance between the WCV and the to-be-charged sensor), or both factors combined through some weights to calculate the priority of the charging tasks. One advantage of applying BP&R is that we can predict, over time, the potential charging needs that may overwhelm the on-demand scheduling and eliminate those incidents beforehand. This is similar to pulling the time factor from on-demand scheduling and putting it into a more global context to concentrate on tackling the scheduling problem considering only the distance factor. Thus, a simple distance-based near job first (NJF) algorithm is sufficient for charge scheduling when time-related factors are considered prior to the determination of the charging order as a result of BP&R mechanism. The resulted scheduling strategy takes advantage of the on-demand feature to roughly determine the power consumption in a network, while also proactively includes additional sensor nodes that could potentially cause a charging bottleneck in the future as long as there is spare WCV battery power and time. The on-demand and proactive features of the proposed strategy ensure that this method can adapt to the dynamic charging needs of sensors, while at the same time taking precautionary measures to avoid possible burst requests at a later time.

The other advantage of applying BR&R mechanism is that we are able to derive a shorter path length of charge scheduling based on the NJF algorithm, which also results in a tighter charging threshold. In particular, in a given case with network sizes of 500 m × 500 m, we demonstrated through simulation that the threshold value could be set as low as 2.9% of the sensors' remaining power. This is significant in that a lower charging threshold corresponds to fewer charging requests and less pressure on charge scheduling. In addition, extensive simulation results demonstrate that when combining NJF or TADP with the proposed BP&R strategy, network lifetime can be improved by up to 210.6% and 166.8%, respectively, comparing to using NJF or TADP alone.

For future research, it would be interesting to apply the proposed algorithm to charge scheduling in WRSNs with multiple mobile chargers or chargers with multiple charging capabilities. Improvements in charging efficiency may be possible in these contexts. In addition, the proposed deadline estimation mechanism requires a node to update its deadline information for recharging when the difference between its newly estimated and current deadline is greater than a fixed pre-defined threshold. In reality, the power consumption of each node may be dynamic, which means that the threshold does not need to be fixed. In the future, we will explore the effects of adjusting this threshold according to the nodes' loading conditions. Finally, instead of selecting node(s)

randomly within the potential bottlenecked window for candidate(s) of proactive charging, we will work on developing strategies for determining specific node(s) that may be more effective than others for overall scheduling performance.

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